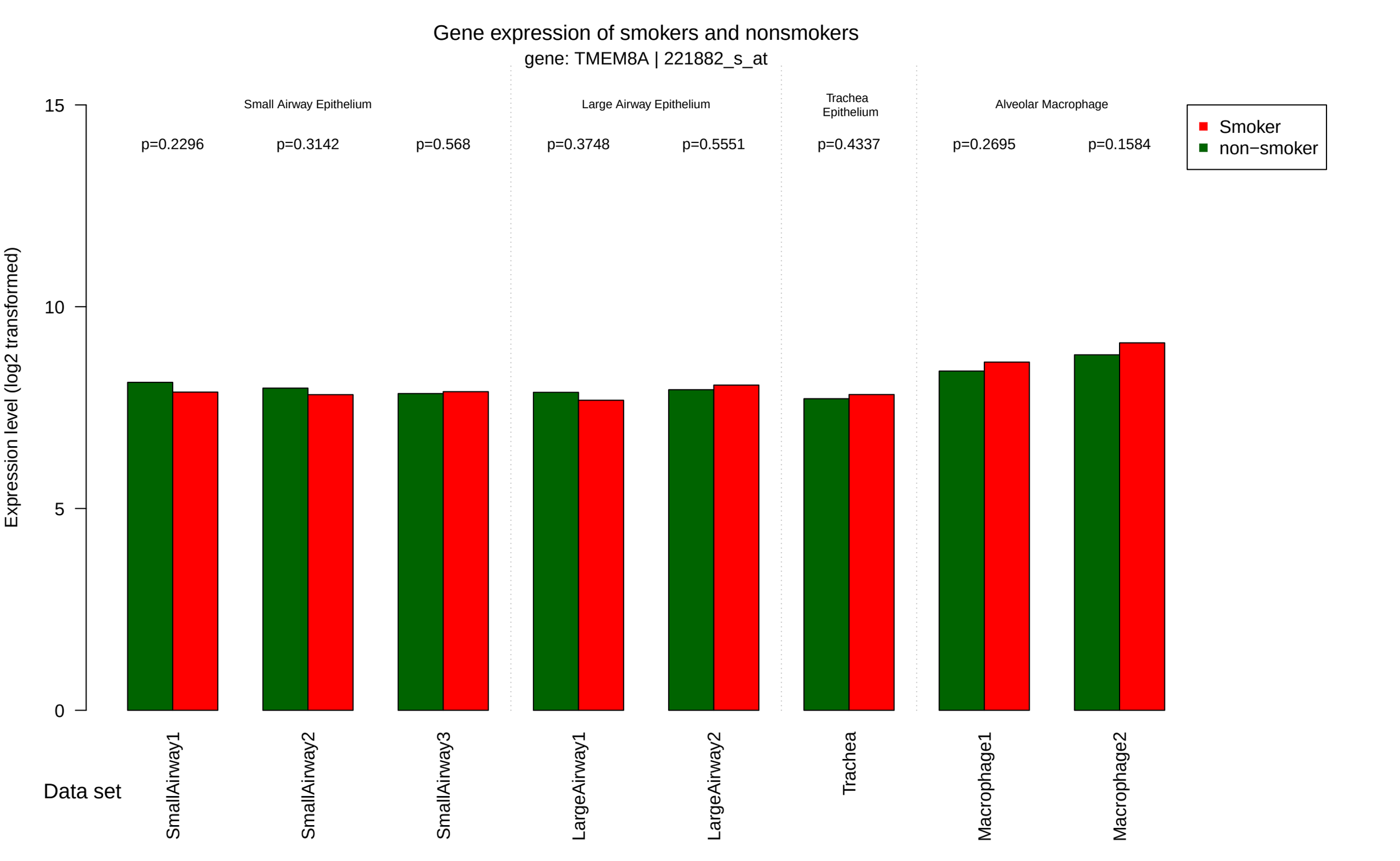
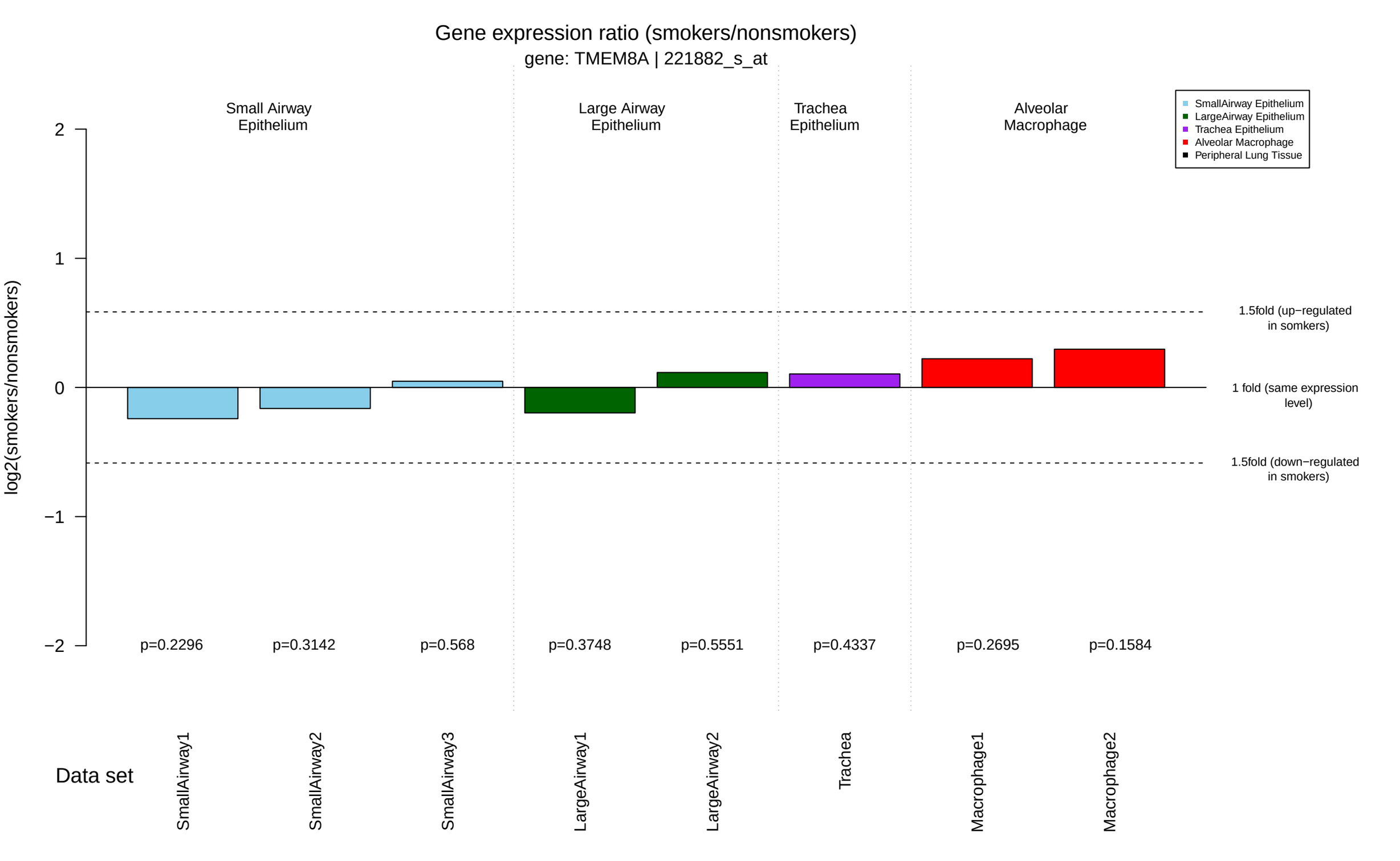
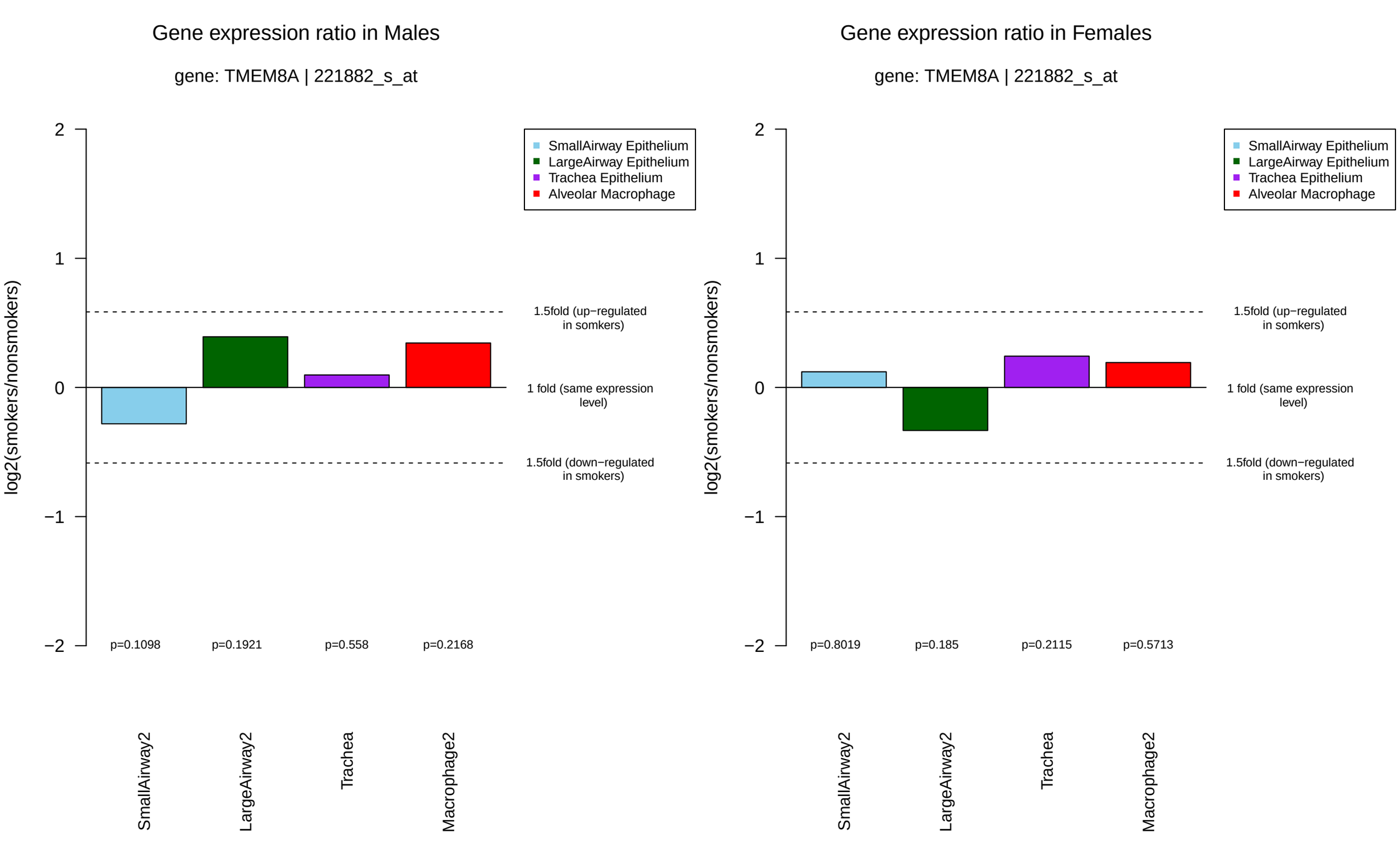
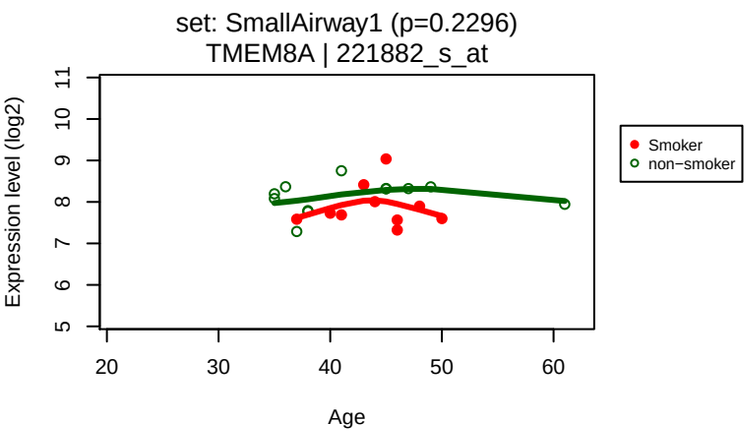
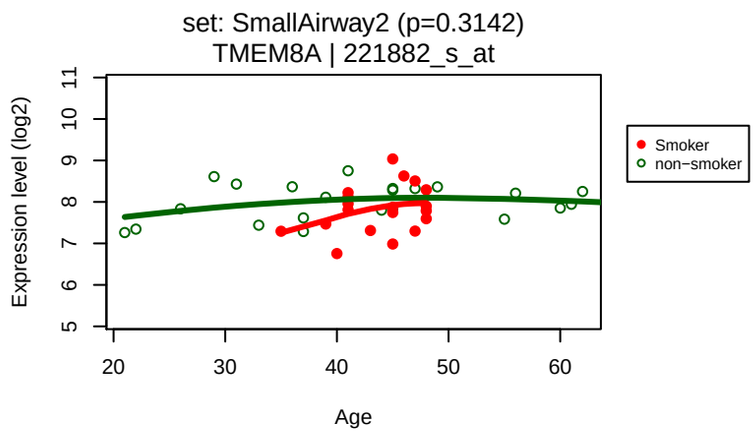
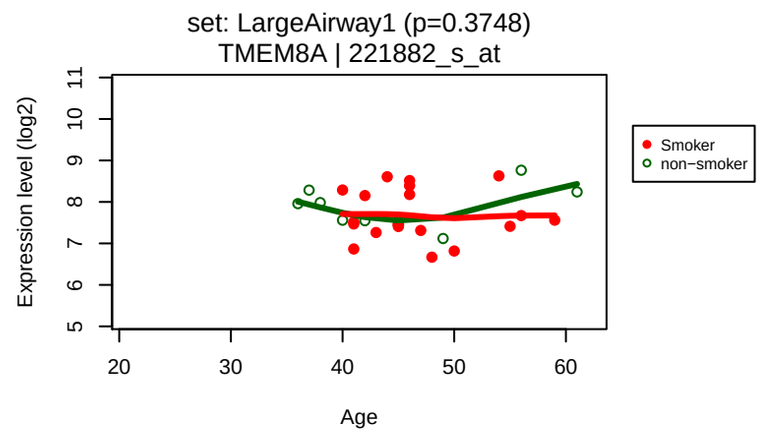

Gene expression of smokers and nonsmokers
gene: TMEM8A | 221882_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2296
p=0.3142
p=0.568
p=0.3748
p=0.5551
p=0.4337
p=0.2695
p=0.1584
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

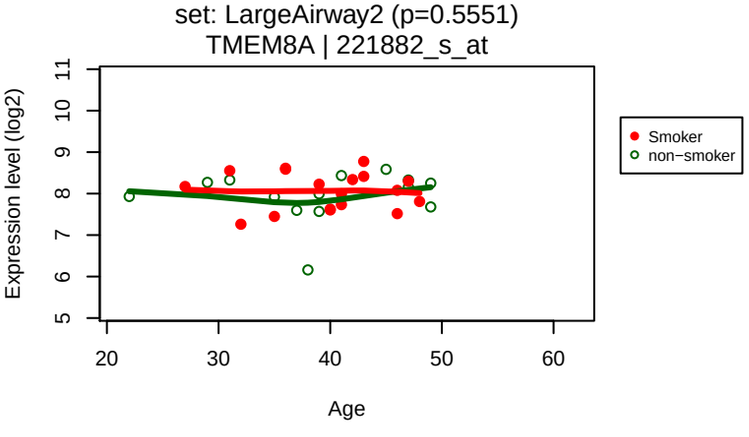

Gene expression ratio (smokers/nonsmokers)
gene: TMEM8A | 221882_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2296
p=0.3142
p=0.568
p=0.3748
p=0.5551
p=0.4337
p=0.2695
p=0.1584
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

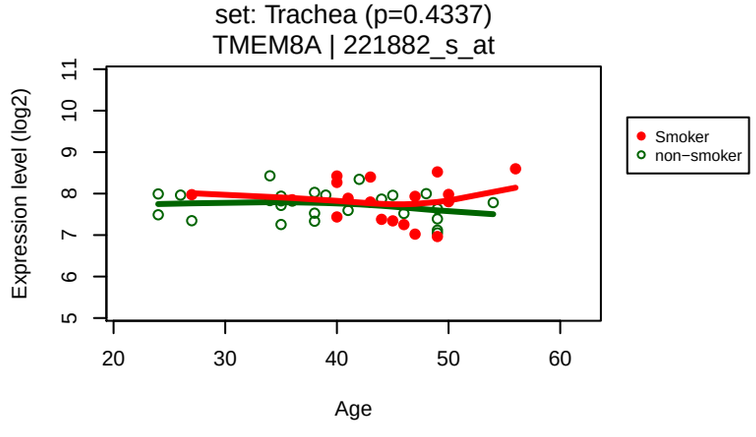

Gene expression ratio in Males
gene: TMEM8A | 221882_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1098
p=0.1921
p=0.558
p=0.2168
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM8A | 221882_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8019
p=0.185
p=0.2115
p=0.5713
SmallAirway2
LargeAirway2
Trachea
Macrophage2

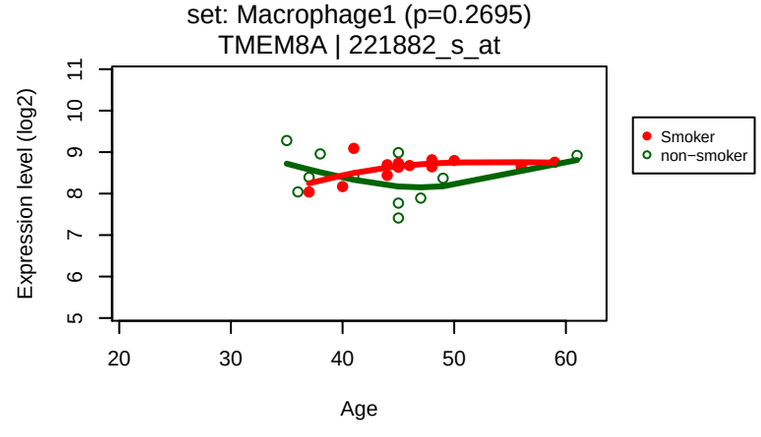

set: SmallAirway1 (p=0.2296)
TMEM8A | 221882_s_at
Expression level (log2)
Age
Smoker
non−smoker

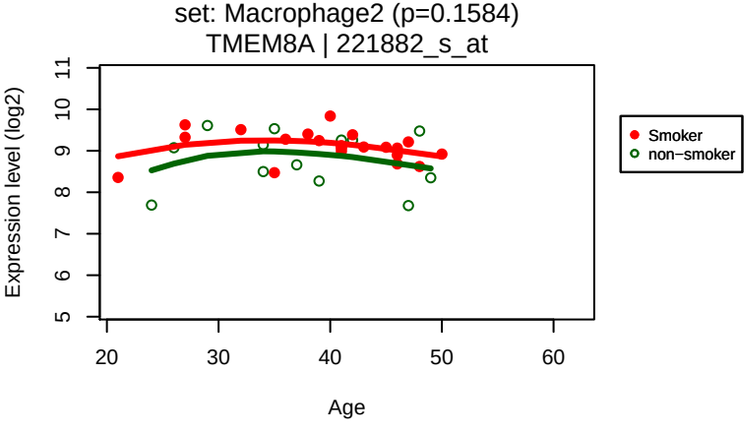

set: SmallAirway2 (p=0.3142)
TMEM8A | 221882_s_at
Expression level (log2)
Age
Smoker
non−smoker

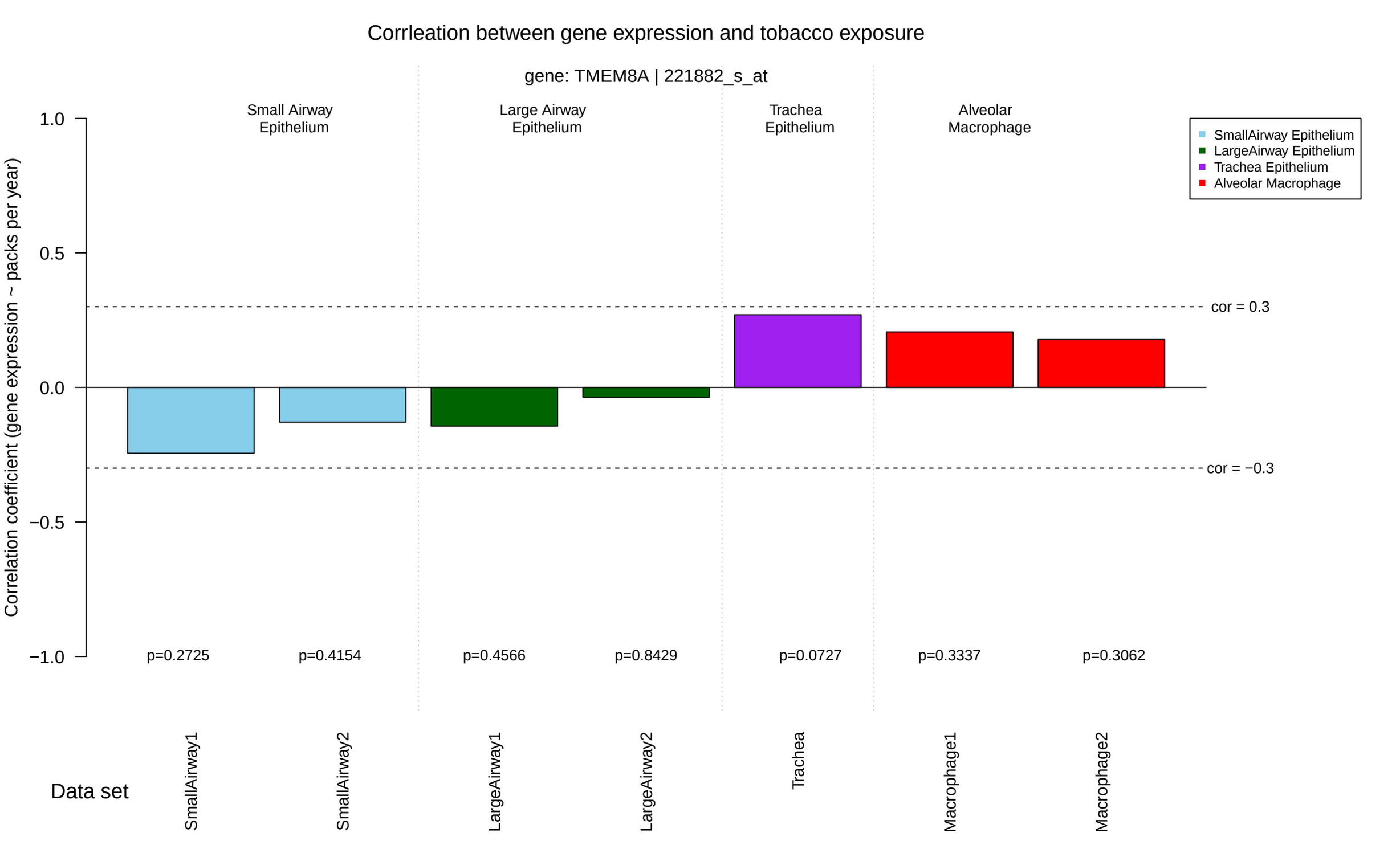

set: LargeAirway1 (p=0.3748)
TMEM8A | 221882_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5551)
TMEM8A | 221882_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4337)
TMEM8A | 221882_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2695)
TMEM8A | 221882_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1584)
TMEM8A | 221882_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM8A | 221882_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2725
p=0.4154
p=0.4566
p=0.8429
p=0.0727
p=0.3337
p=0.3062
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2